

Guidance for MTH1004 Term 1 Coursework “Report 1”

This document contains additional guidance for the MTH1004 Term 1 coursework.

Creating a pdf file of your poster

There are many different word processors and text editors that can produce pdf files. Below is a solution that uses PowerPoint that is part of the MS Office Suite that all students have access to. It can be run in the web browser, so doesn't require installation of any software on your device and should work for all students. (You do not have to follow these instructions to create your pdf.)

1. To open PowerPoint in your browser go to <https://www.microsoft365.com/launch/powerpoint> You might have to log in with your university username and password.
2. Click on the “Blank Presentation” button.
3. Use Design > Slide Size > Custom Slide Size to set the size to the required A4 format.
4. Use Insert > Text Box and Insert > Picture to add and organise your content on the slide.
5. Choose File > Save as > Download as PDF
6. Open the downloaded pdf file to check that it looks as desired.

Why pdf? The pdf file format is specifically designed to look exactly the same on every device. This cannot be said about other types of files like .ppt or .docx. A poster in .docx format might look good on your computer but could look completely different when I open it on mine.

Design tips for scientific poster presentations

- Design is subjective, but there are simple guidelines that help create an aesthetic but clear data analysis poster.
- Use a common colour theme, avoiding an excessive number of different colours, font types, sizes, etc. Use at most 2 different font types, one for title and headings and one for text, with titles and headings possibly in bold.
- Retain margins and white space for a clean look but avoid excessively large, blank areas.
- When resizing an image to fit onto the page, avoid distortions (squeezing and stretching) and make sure the figure is big enough so that all axis labels and annotations are legible.
- Try to align text boxes and images.
- Add a caption under your figure and reference it in the text, e.g. “Figure 1 illustrates the relationship between X and Y.” Include units of measurements in your axis labels.
- Write in full sentences using concise, neutral language. Avoid informal and conversational language. Take care in wording your conclusions (e.g. “these results suggest”, “the reason might be”) and avoid hyperbolic language (e.g. “without a shadow of a doubt”, “massively”, “extremely”, “significantly”).
- For more guidance on academic writing, referencing, designing posters, and more, please visit the [University of Exeter Study Zone](#). You can also refer to the book, *Communicating with Data: The Art of Writing for Data Science* by Deborah Nolan and Sara Stoudt, which is available through the university library.

We do not prescribe any specific style and layout for your poster and look forward to seeing your creative ideas. However, in anticipation of questions along those lines, I would say that the following layouts are all acceptable:



R code

Try your best to make your R code readable for someone who is reviewing or trying to understand your code (such your assessor or your future self). Use indentation, white space, blank lines and comments to structure your code and improve readability. Consider the following specific guidelines:

- Save your script file with filename extension .R or .r and use a descriptive filename (e.g. coursework-1-code.R rather than script.R).
- Use spaces around mathematical operators (+, =, - etc).
- Place a space after every comma.
- Try to stick to about 80 characters per code line and use line breaks and indentation to break a long line up into several shorter ones. I recommend going to a new line every time you use the pipe operator (%>%) or add a layer to your plot (i.e., +) (RStudio helps you with indentation.)
- Separate unrelated code chunks by blank lines (like paragraphs in a written text) and briefly describe each code chunk with a comment.
- Save your graph(s) as .png files instead of .pdf so they appear clearer in your poster.
- Stick to functions you learned in the lectures as much as possible. If you need to use a function not covered in the module, please add a comment that indicates where you learned it.